

JRAHS 2006 TRIAL HSC - EXT I

Question 1.	Marks
(a) Solve for x : $\frac{1}{x-2} \geq 2$.	3
(b) Find: $\lim_{h \rightarrow 0} \left(\frac{\cos 2h - 1}{h} \right)$.	2
(c) The point P divides $A(-1, 5)$ and $B(3, -2)$ in the ratio $r : 1$.	
(i) Find the coordinates of P in terms of r .	2
(ii) Find the value of r when the line $2x - 3y + 4 = 0$ intersects the interval AB .	2
(d) Evaluate $\int_0^1 (x^2 + 1)^3 dx$.	3

Question 2. **[START A NEW PAGE]**

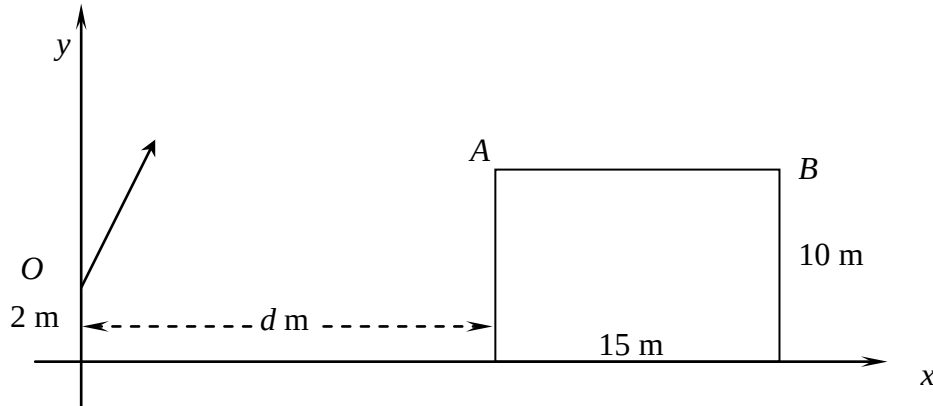
- (a) A plate is initially heated to 55°C , and it then cools to 41°C in 10 minutes. If the surrounding temperature, $S^{\circ} \text{C}$, is 22°C and assuming Newton's Law of Cooling:
- $$\frac{dT}{dt} = -k(T - S).$$
- (i) Find the temperature of the plate 25 minutes from the start of cooling (to 1 decimal place). **3**
- (ii) Find the time for the plate to cool to 25°C (to 1 decimal place). **2**
- (iii) Sketch the graph of the rate of temperature, $\frac{dT}{dt}$, versus the temperature T . **1**
- (b) The displacement x metres of a particle after t seconds, is given by:
- $$x = 5 \sin 3t - 7 \cos 3t.$$
- (i) Show that the motion of the particle is SHM. **2**
- (ii) Find the maximum displacement. **1**
- (iii) Find the time when the particle first passes through the centre of motion (correct to 1 decimal place). **2**
- (iv) Sketch the graph of the acceleration $\ddot{x}$ versus displacement x . **1**

Question 3.**[START A NEW PAGE]****Marks**

- (a) Differentiate $\cos^{-1}\left(-\frac{1}{x}\right)$ with respect to x . Answer in simplified form. **3**
- (b) (i) On the same set of axes, sketch the graphs of $y = \sin^{-1} x$ and $y = \tan^{-1} x$. **2**
- (ii) Given that: $\int_0^1 \sin^{-1} x dx = \frac{\pi}{2} - 1$, find the area of the region bounded by $y = \sin^{-1} x$, $y = \tan^{-1} x$ and $x = 1$. **3**
- (c) (i) Show that $y = e^{-x} \sin 2x$ is a solution to the differential equation: $y'' + 2y' + 5y = 0$. **3**
- (ii) Hence, or otherwise, find $\int e^{-x} \sin 2x dx$. **1**

Question 4.**[START A NEW PAGE]**

- (a) A fire truck arrives at a burning building 10 metres high and 15 metres wide. The water nozzle hose on the fire truck is 2 metres above the ground and d metres from the building, as shown in the diagram.



The angle of elevation of the hose, α , can be adjusted to range from 10° to 45° . The parametric equations for the water particles from the nozzle are given by: $x = 30t \cos \alpha$ and $y = 30t \sin \alpha - 5t^2$, where t is the time in seconds when $g = 10$.

- (i) Show that the trajectory path of the water is given by the equation: **1**
- $$y = x \tan \alpha - \frac{x^2}{180}(1 + \tan^2 \alpha).$$
- (ii) The hose nozzle is adjusted to an angle of elevation of 45° . **2**
- Find the distance, d , from the building if the water is to reach the furthest point B on top of the building as shown (answer to the nearest centimetre).

Q 4 continues over the page

Q 4 part (a) continued

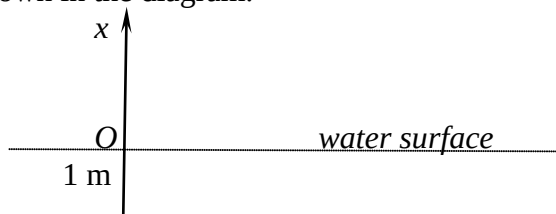
Marks

- (iii) Find the angle of elevation α of the nozzle, for the water to reach position A, when the hose nozzle is 20 metres from the burning building (answer to nearest minute). 2
- (b) Find $\int \frac{4x-7}{2x^2+1} dx$. 3
- (c) (i) For $t > 0$, find the limiting sum of: $e^{-t} + e^{-2t} + e^{-3t} + \dots$ 1
- (ii) Hence, find an expression for the series; $e^{-t} + 2e^{-2t} + 3e^{-3t} + \dots$ 1
- (d) A semi-circle of radius r has the equation: $y = \sqrt{r^2 - x^2}$.
- (i) Find $\frac{dy}{dx}$ at the point $P(x, y)$. 1
- (ii) Prove that the tangent, at any point P on the semi-circle, is perpendicular to the radius. 1

Question 5.

[START A NEW PAGE]

- (a) Find the greatest coefficient in the expansion of $(4x+5)^{11}$. 3
(Leave the answer in index form).
- (b) A ping pong ball is initially placed 1 metre beneath the surface of the water, as shown in the diagram.



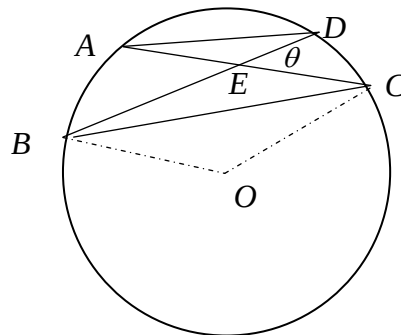
The ping pong ball is released in the water with an acceleration of $\ddot{x}$ m/s², where $\ddot{x} = -625x$, and where x metres is the displacement of the motion measured from the water surface.

- (i) Is the motion of the ping pong ball only SHM? Give reasons. 1
- (ii) Prove that: $\frac{d}{dx} \left(\frac{v^2}{2} \right) = \ddot{x}$. 2
- (iii) Find the expression for the ping pong ball's velocity v m/s when it is in the water. 2
- (iv) Find the velocity of the ball at the water's surface. 1
- (v) Assuming there is no air resistance and the acceleration due to gravity is 10 m/s², derive an expression for the displacement in air in terms of v 2
- (vi) Find the maximum height that the ping pong ball reaches above the surface of the water. 1

Question 6.	[START A NEW PAGE]	Marks
(a)	How many groups of 2 men and 2 women can be chosen from 6 men and 8 women?	2
(b)	Six letter words are formed from the letters of the word CYCLIC .	
(i)	How many different 6-letter words can be formed?	2
(ii)	How many 6 letter words can be formed, if no 'C's are together?	2
(iii)	What is the probability of all the 'C's together, if it is known a vowel is at the end?	2
(c)	Prove, by the method of mathematical induction that: $\sin q + \sin 3q + \sin 5q + \dots + \sin(2n-1)q = \frac{1 - \cos 2nq}{2 \sin q}, \text{ for } n = 1, 2, 3, \dots$	4

Question 7. [START A NEW PAGE]

- (a) At the end of each month, for 15 years, a man invests \$400 at an interest rate which is paid monthly at 6% *pa*.
- (i) Show that the value of his first payment, at the end of 15 years, is \$976.75
- (ii) Find the value of the man's total investment at the end of the 15 years.
- (b) A circle, centre O with a constant radius r , is such that the chords AC and BD intersect at point E , $\angle CED = \theta$ radians and $\angle BOC = \frac{2\pi}{3}$ radians, as shown the diagram.



Not to scale

- (i) Show that the sum of the arcs AB and CD equal $2r\theta$, give reasons.
- (ii) Show that the perimeter P of the shape $ABCD$, where BC, AD are chords and CD, AB are arc lengths, is given by:
- $$P = r \left(2\theta + \sqrt{3} + 2 \sin \left(\frac{\pi}{3} - \theta \right) \right).$$
- (iii) Find the value of θ , in the domain $0 \leq \theta \leq \frac{\pi}{2}$ for the perimeter of $ABCD$ to have a maximum value. Justify your answer.

(a) $\frac{1}{x-2} \geq 2$ $x \neq 2$
 $(x-2) \geq 2(x-2)^2$

$(x-2) - 2(x-2)^2 \geq 0$

$(x-2)(1-2(x-2)) \geq 0$

$(x-2)(5-x) \geq 0$

$\therefore$ Solution $2 \leq x \leq 2\frac{1}{2}$

b) $\lim_{h \rightarrow 0} \left(\frac{\cos 3h - 1}{h} \right) \left(\frac{\cos 3h + 1}{\cos 3h + 1} \right)$

$= \lim_{h \rightarrow 0} \frac{\cos^2 3h - 1}{h [\cos 3h + 1]}$

$= \lim_{h \rightarrow 0} \frac{-\sin^2 3h}{h [\cos 3h + 1]}$

$= \lim_{h \rightarrow 0} \frac{\sin 3h}{3h} \cdot \frac{-3 \sin 3h}{\cos 3h + 1}$

$= 1 \cdot \frac{0}{2}$

$= 0$

c) (i) $A(-1, 5)$ $B(3, -2)$

$P \equiv \left[\frac{3r-1}{r+1}, \frac{-2r+5}{r+1} \right]$

(ii) $2 \left[\frac{3r-1}{r+1} \right] - 3 \left[\frac{-2r+5}{r+1} \right] + 4 = 0$

$6r-2 + 6r-15 + 4r+4 = 0$
 $16r = 13$

$r = \frac{13}{16}$

(a) $\int_0^1 (x^2+1)^3 dx$
 $= \int_0^1 (x^6 + 3x^4 + 3x^2 + 1) dx$
 $= \left[\frac{x^7}{7} + \frac{3}{5}x^5 + x^3 + x \right]_0^1$
 $= 2 \frac{26}{35}$

2 $\frac{dT}{dt} = -k(T-T_0)$

$T = T_0 + A e^{-kt}$

$T_0 = 22^\circ$ And $t=0$ $T=55$

$55 = 22 + A e^0$

$A = 33 \Rightarrow T = 22 + 33 e^{-kt}$

And $41 = 22 + 33 e^{-10k}$

$19 = 33 e^{-10k}$

$k = \frac{1}{10} \ln \frac{33}{19} = \frac{1}{10} \ln \left(\frac{33}{19} \right)$

$T = 22 + 33 e^{-kt}$

(i) $t = 25$ $\frac{-25}{10} \ln \frac{33}{19}$

$T = 22 + 33 e^{-kt}$

$= 30.3^\circ \text{C}$ $\frac{-t}{10} \ln \frac{33}{19}$

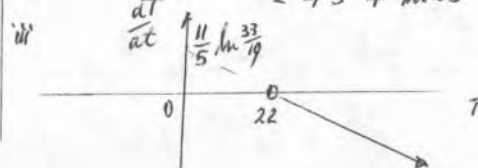
$25 = 22 + 33 e^{-kt}$

$\frac{-t}{10} \ln \frac{33}{19} = \frac{3}{33}$

$\frac{-t}{10} \ln \frac{33}{19} = -\ln 11$

$t = \frac{10 \ln 11}{\ln \frac{33}{19}}$

$= 43.4 \text{ mins}$



(3)

2(b) $x = 5 \sin 3t - 7 \cos 3t$

i) $\dot{x} = 15 \cos 3t + 21 \sin 3t$

$\ddot{x} = -45 \sin 3t + 63 \cos 3t$

$= -9 [5 \sin 3t - 7 \cos 3t]$

$\ddot{x} = -9x$

which is of the form $\ddot{x} = -n^2(x-h)$
 $n=3$ $h=0$

$\therefore$ motion SHM.

ii) Max displacement $= \sqrt{5^2 + 7^2}$
 $= \sqrt{25 + 49}$
 $= \sqrt{74} \text{ units.}$

Max velocity $= \sqrt{15^2 + 21^2}$
 $= 3\sqrt{74} \text{ m/s}$

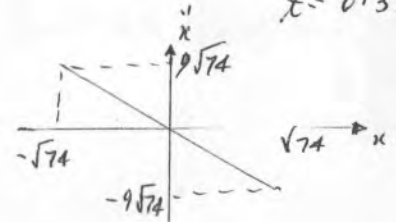
iii)

$x=0$ $5 \sin 3t - 7 \cos 3t = 0$

$\tan 3t = \frac{7}{5}$

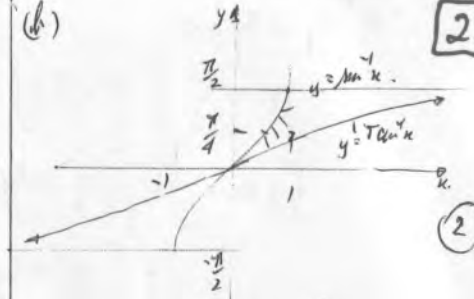
$3t = \tan^{-1} \frac{7}{5}$

$t = 0.322 \text{ s}$



iv)

5 (a) $\frac{d}{dx} \cos^{-1} \left(\frac{1}{x} \right) = \frac{1}{x^2} \cdot \frac{-1}{\sqrt{1 - \frac{1}{x^2}}}$
 $= \frac{-\sqrt{x^2}}{x^2 \sqrt{x^2 - 1}}$
 $= \frac{-|x|}{x^2 \sqrt{x^2 - 1}}$
 $= \frac{-1}{|x| \sqrt{x^2 - 1}}$



(b) $\int_0^1 \tan^{-1} x dx = \frac{\pi}{4} - \int_0^{\pi/4} \tan y dy$
 $= \frac{\pi}{4} - \int_0^{\pi/4} \frac{\sin y}{\cos y} dy$
 $= \frac{\pi}{4} + \left[\ln \cos y \right]_0^{\pi/4}$
 $= \frac{\pi}{4} + \ln \left(\frac{1}{\sqrt{2}} \right)$
 $= \frac{\pi}{4} - \frac{1}{2} \ln 2$

i. Region $\int_0^1 \sin^{-1} x dx = \int_0^{\pi/2} \tan^{-1} u du$
 $= \frac{\pi}{2} - 1 - \left[\frac{\pi}{4} - \frac{1}{2} \ln 2 \right]$
 $= \frac{\pi}{4} - 1 + \frac{1}{2} \ln 2$

c) (i) $y = e^{-x} \sin 2x$
 $\frac{dy}{dx} = -e^{-x} \sin 2x + 2e^{-x} \cos 2x$
 $= e^{-x} [2 \cos 2x - \sin 2x]$
 $\frac{d^2 y}{dx^2} = -e^{-x} [2 \cos 2x - \sin 2x] + e^{-x} [-4 \sin 2x - 2 \cos 2x]$
 $= e^{-x} [-3 \sin 2x - 4 \cos 2x]$
 $\therefore y'' + 4y' + 5y = e^{-x} [-3 \sin 2x - 4 \cos 2x + 4 \cos 2x - 2 \sin 2x + 5 \sin 2x]$
 $= 0$

4

i) $y'' + 2y' + 5y = 0$

$$y' = -\frac{1}{5} [y'' + 2y']$$

$$\int y' dx = -\frac{1}{5} \int (y'' + 2y') dx$$

$$= -\frac{1}{5} [y' + 2y] + C$$

$$\int e^{-x} \sin 2x dx = -\frac{1}{5} \left[e^{-x} (2 \cos 2x - \sin 2x) + 2e^{-x} \sin 2x \right] + C$$

$$= -\frac{e^{-x}}{5} [2 \cos 2x + \sin 2x] + C \quad (1)$$

4(c) (i) $x = 30t \cos \alpha$

$y = -5x^2 + 30x \tan \alpha$

$t = \frac{x}{30 \cos \alpha}$

$y = -5 \left(\frac{x}{30 \cos \alpha} \right)^2 + 30 \tan \alpha \frac{x}{30 \cos \alpha}$

$y = \frac{-x^2}{180} \sec^2 \alpha + x \tan \alpha$

or $y = \frac{-x^2}{180} [1 + \tan^2 \alpha] + x \tan \alpha \quad (1)$

i) $\alpha = 45^\circ$

$y = 8$

$x = d + 15$

$\therefore 8 = \frac{-x^2}{180} (1 + 1) + x$

$x^2 - 90x + 720 = 0$

$x = \frac{90 \pm \sqrt{90^2 - 4 \times 720}}{2}$

$d + 15 = 81.12 \text{ or } 17.75$

$d = 66.12 \text{m}$ furthest distance.

(2)

3

iii) $A(20, 8)$

$8 = -\frac{400}{180} [1 + \tan^2 \alpha] + 20 \tan \alpha$

$72 = -20 [1 + \tan^2 \alpha] + 180 \tan \alpha$

$20 \tan^2 \alpha - 180 \tan \alpha + 92 = 0$

$5 \tan^2 \alpha - 45 \tan \alpha + 23 = 0$

$\tan \alpha = \frac{45 \pm \sqrt{45^2 - 4 \times 5 \times 23}}{2 \times 5}$

$= 0.544 \text{ or } 8.46$

Angle elevation

$\alpha = 28^\circ 33' \text{ as } 0 \leq \alpha \leq 45^\circ$

(2)

(b) $\int \frac{4x-7}{2x^2+1} dx = \int \left(\frac{4x}{2x^2+1} - \frac{7}{2x^2+1} \right) dx$

$= \int \left(\frac{4x}{2x^2+1} - \frac{7}{2} \cdot \frac{1}{x^2 + \frac{1}{2}} \right) dx$

$= \ln(2x^2+1) - \frac{7}{2} \cdot \frac{1}{\sqrt{\frac{1}{2}}} \tan^{-1} \sqrt{\frac{1}{2}} + C$

$= \ln(2x^2+1) - \frac{7}{\sqrt{2}} \tan^{-1}(x\sqrt{2}) + C$

(2)

(c) (i) $e^{-t} + e^{-2t} + e^{-3t} + \dots = \frac{e^{-t}}{1 - e^{-t}} \text{ as } |e^{-t}| < 1 \text{ for } t > 0$

$= \frac{1}{e^t - 1}$

(1)

(ii)

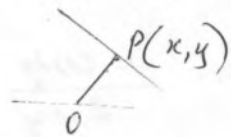
Now $\frac{d}{dt} [e^{-t} + e^{-2t} + e^{-3t} + \dots] = \frac{d}{dt} (e^t - 1)^{-1}$

$= -e^{-t} - 2e^{-2t} - 3e^{-3t} + \dots = - (e^t - 1)^{-2} \cdot e^t$

$\therefore e^{-t} + 2e^{-2t} + 3e^{-3t} + \dots = \frac{e^t}{1 - e^{-t}}$

(1)

(d) $y = \sqrt{r^2 - x^2}$
 Gradient $m_1 = \frac{dy}{dx} = \frac{-x}{\sqrt{r^2 - x^2}}$



(1)

Gradient OP = $\frac{y-0}{x-0}$
 $= \frac{y}{x}$
 $m_2 = \frac{\sqrt{r^2 - x^2}}{x}$

$\therefore m_1 \times m_2 = \frac{-x}{\sqrt{r^2 - x^2}} \cdot \frac{\sqrt{r^2 - x^2}}{x}$
 $= -1$

$\therefore$ Tangent $\perp$ radius.

5 $(a+b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$

$\therefore T_{r+1} = \binom{11}{r} (4x)^{11-r} 5^r$

$T_r = \binom{11}{r-1} (4x)^{12-r} 5^{r-1}$

$\frac{T_{r+1}}{T_r} = \frac{11!}{r!(11-r)!} \cdot \frac{(r-1)!(12-r)!}{11!} \cdot \frac{4^{11-r}}{4^{12-r}} \cdot \frac{5^r}{5^{r-1}} \cdot \frac{x^{11-r}}{x^{12-r}}$
 $= \frac{12-r}{r} \cdot \frac{5}{4} \cdot \frac{1}{x}$

For largest coefficient $\frac{5(12-r)}{4r} \geq 1$

$60 - 5r \geq 4r$
 $9r \geq 60$
 $r \geq 7$

$\therefore$ Largest coefficient $\binom{11}{6} 4^5 5^6$

(1)

(1)

(1)

5

(b) (i) No motion does not oscillate.

(ii) $\frac{d}{dx} \left(\frac{v^2}{2} \right) = \frac{d}{dv} \left(\frac{v^2}{2} \right) \cdot \frac{dv}{dx} v$
 $= \frac{2v}{2} \cdot \frac{dv}{dx}$
 $= v \frac{dv}{dx}$
 $= \frac{dx}{dt} \cdot \frac{dv}{dx}$
 $= \frac{dv}{dt}$
 $= \ddot{x}$

2

(iii)

$\ddot{x} = -625x$

$\frac{d}{dx} \left(\frac{v^2}{2} \right) = -625x$

$\frac{v^2}{2} = -625 \frac{x^2}{2} + C$ but $v=0$ $x=-1$

$\therefore 0 = -\frac{625}{2} + C$

$C = +\frac{625}{2}$

$\therefore \frac{v^2}{2} = \frac{625}{2} - 625 \frac{x^2}{2}$

$v^2 = 625(1-x^2)$

$v = 25\sqrt{1-x^2}$ $v > 0$.

2

(iv) At surface $x=0$ $\therefore v=25$ m/s

(v) $\frac{d^2x}{dt^2} = -g$
 $\frac{d}{dx} \left(\frac{v^2}{2} \right) = -10$

1

$$\frac{v^2}{2} = -10x + C \quad x \geq 0 \quad v \leq 25$$

$$\therefore C = \frac{25^2}{2}$$

$$\frac{v^2}{2} = -10x + \frac{625}{2}$$

$$v^2 = -20x + 625$$

$$x = \frac{625 - v^2}{20}$$

(2)

(vi) Max height $v=0$

$$x = \frac{625}{20}$$

$$= 31.25 \text{ m.}$$

(1)

$$(a) \text{ Ways} = \binom{6}{2} \binom{8}{2} = 420$$

(2)

$$(b) (i) \text{ Number words } \frac{6!}{3!} = 120$$

(2)

$$(ii) \begin{array}{l} C \Delta C \Delta C \Delta \\ C \Delta \Delta C \Delta C \\ C \Delta C \Delta \Delta C \end{array}$$

$$\text{Total} = 4 \times 3! = 24$$

(2)

$$(iii) \text{ (ccc) Y L I or I (ccc) Y L}$$

$$\text{Ways I} = 2!$$

$$\text{Ways C} = 1$$

$$\text{Total} = 2 \times 3!$$

$$\text{I end} = 12$$

$$\text{Ways [cccYL]} = \frac{5!}{3!} = 20$$

$$\text{Total} = 2 \times 20 = 40$$

$$\text{Probability (If end is I)} = \frac{12}{40} = \frac{3}{10}$$

(2)

[7]

(b) Step 1 $n=1$

$$\text{LHS} = \sin q$$

$$\text{RHS} = \frac{1 - \cos 2q}{2 \sin q}$$

$$= \frac{1 - (1 - 2 \sin^2 q)}{2 \sin q}$$

$$= \frac{2 \sin^2 q}{2 \sin q}$$

$$= \sin q$$

$$= \text{LHS}$$

(1)

Step 2

i. True $n=1$.

Assume true $n=k$ $\sin q + \sin 3q + \dots$

$$\sin(2k-1)q = \frac{1 - \cos 2kq}{2 \sin q}$$

To prove true $n=k+1$ $\sin q + \sin 3q + \dots$

$$\sin(2k+1)q = \frac{1 - \cos 2(k+1)q}{2 \sin q}$$

LHS = $\sin q + \sin 3q + \dots + \sin(2k-1)q + \sin(2k+1)q$

$$= \frac{1 - \cos 2kq}{2 \sin q} + \sin(2k+1)q \quad (\text{By assumption})$$

$$= \frac{1 - \cos 2kq}{2 \sin q} + 2 \sin q \sin(2k+1)q$$

(1)

$$= \frac{1 - \cos[(2k+1)q - q] + 2 \sin q \sin(2k+1)q}{2 \sin q}$$

$$= \frac{1 - \{ \cos(2k+1)q \cos q + \sin(2k+1)q \sin q \} + 2 \sin q \sin(2k+1)q}{2 \sin q}$$

$$= \frac{1 - \{ \cos(2k+1)q \cos q - \sin q \sin(2k+1)q \}}{2 \sin q}$$

$$= \frac{1 - \cos[(2k+1)q + q]}{2 \sin q}$$

(1)

$$= \frac{1 - \cos 2(k+1)q}{2 \sin q}$$

$\therefore$ If statement true $n=k$ it is also true $n=k+1$.

Since true for $n=1$ it is also true for $n=1+1=2, n=2+1=3$ and so on

i) 

$$\frac{dP}{d\theta} = r \left[2 - 2 \cos \left(\frac{\pi}{3} - \theta \right) \right]$$

(11)

$$\frac{d^2P}{d\theta^2} = 2r \sin \left(\frac{\pi}{3} - \theta \right)$$

For maximum perimeter $\frac{dP}{d\theta} = 0$

$$r \left[2 - 2 \cos \left(\frac{\pi}{3} - \theta \right) \right] = 0$$

$$r \neq 0 \quad \cos \left(\frac{\pi}{3} - \theta \right) = 1$$

$$\frac{\pi}{3} - \theta = 0 \quad \text{for } 0 < \theta < \frac{\pi}{2}$$

$$\therefore \theta = \frac{\pi}{3}$$

For nature test $\frac{d^2P}{d\theta^2}$ for concavity

$$\text{at } \theta = \frac{\pi}{3} \quad \frac{d^2P}{d\theta^2} = r \times 0 = 0$$

Test gradients:

θ	1	$\frac{\pi}{3}$	1.1
$\frac{dP}{d\theta}$	2×10^{-3}	0	2×10^{-3}

gradients same sign / - /

$\therefore$ Inflection point at $\theta = \frac{\pi}{3}$ and monotonic increasing, continuous for $0 < \theta < \frac{\pi}{2}$

$\therefore$ Maximum perimeter at end points of domain
i.e. $\theta = \frac{\pi}{2}$